

Title of the Presentation

SUBTITLE OR CONFERENCE NAME

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Outline

Introduction

Background & Related Work

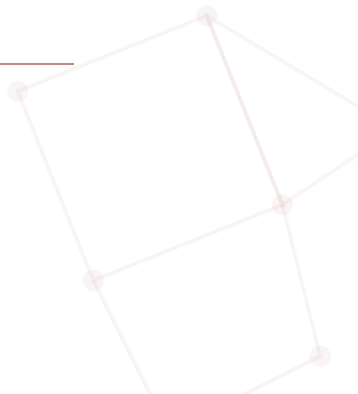
Proposed Methodology

Experiments & Results

Conclusion



Introduction



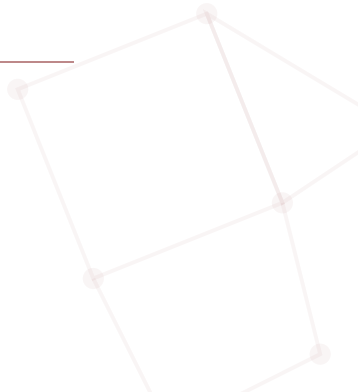
The Problem

- Describe the core problem you are addressing.
- Why is this problem important in AI/ML?
- What are the limitations of current approaches?

Our Contribution

- Briefly state your proposed solution or main findings.
- Highlight key improvements (e.g., accuracy, efficiency).

Background & Related Work

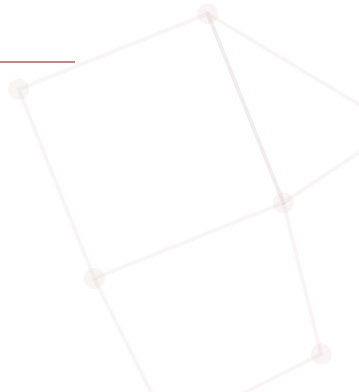


- ☰ **Approach A [Smith et al., 2021]:** Solves X but struggles with Y.
- ☰ **Approach B [Jones et al., 2022]:** Improves Y using technique Z, but is computationally expensive.

🔍 Gap in the Literature

There is a need for a model that achieves high accuracy while remaining computationally efficient on edge devices.

Proposed Methodology



Methodology Overview

- ⚙️ Provide a high-level overview of your model architecture or algorithm.
- 🔧 Use modern diagrams with shadow and clean edges.



Figure 1: High-level architecture of the proposed approach.

We optimize the parameters θ by minimizing the empirical risk.

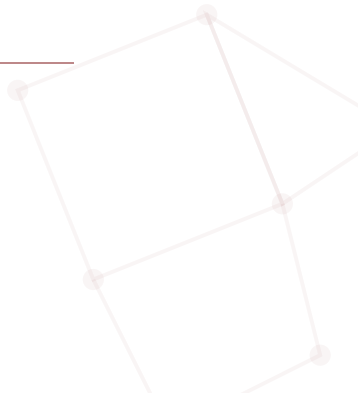
Objective Function

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathbf{x}_i), y_i) + \lambda \mathcal{R}(\theta) \quad (1)$$

where:

- $\ell(\cdot, \cdot)$ is the loss function (e.g., **Cross-Entropy**).
- $\mathcal{R}(\theta)$ is the regularization term to prevent overfitting.
- λ is the regularization hyperparameter.

Experiments & Results



Datasets

- ImageNet-1K
- COCO 2017
- Custom Dataset

Hardware & Details

- 4x NVIDIA A100 GPUs
- PyTorch 2.0
- AdamW ($\text{lr} = 10^{-4}$)

Results: Quantitative Comparison

Table 1: Performance comparison on the benchmark dataset.

Method	Accuracy (%)	F1-Score	Params (M)
Baseline 1	85.2	0.84	25.4
Baseline 2	88.7	0.88	42.1
Ours	91.4	0.91	18.5

🏆 Our model outperforms all baselines while requiring 27% fewer parameters!

Results: Visualizations

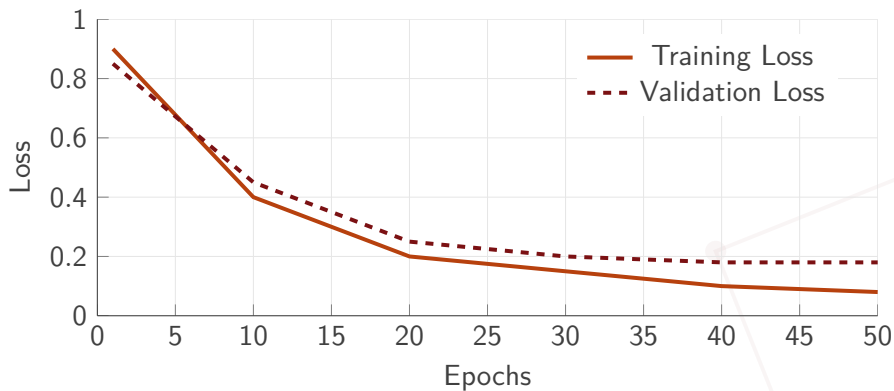


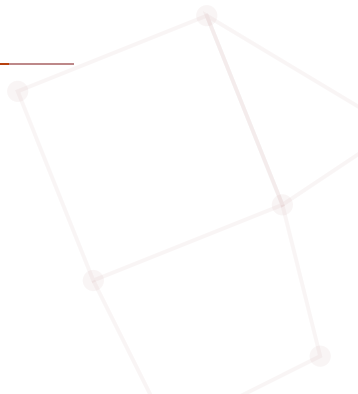
Figure 2: Training and validation loss curve over 50 epochs.



Key Takeaway

Our method achieves **state-of-the-art** performance with **27% fewer parameters**, making it practical for real-world edge deployment.

Conclusion



✓ Summary

- We proposed a novel, lightweight approach for [Task].
- Achieved state-of-the-art results on [Dataset].
- Reduced computational overhead by 30%, enabling edge deployment.

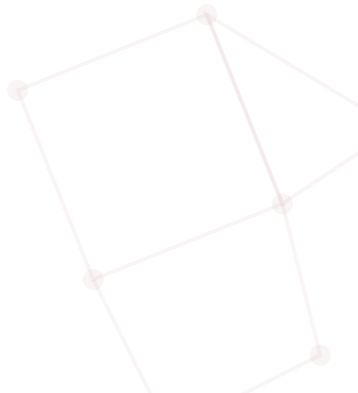
🚀 Future Directions

- Extend the framework to handle multimodal inputs (text + image).
- Investigate theoretical bounds of the optimization landscape.



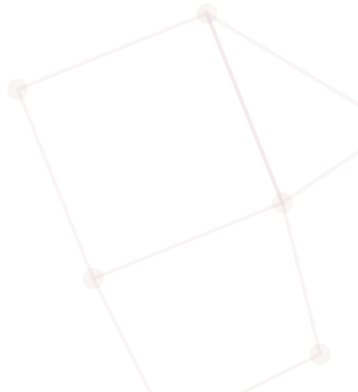
Thank You!

Questions & Discussion



Appendix

Backup Slides



Ablation Study

Table 2: Ablation study on key components of our model.

Configuration	Accuracy (%)	Δ
Full Model	91.4	—
w/o Component A	89.1	−2.3
w/o Component B	87.5	−3.9
w/o Component A & B	85.0	−6.4

Hyperparameter Sensitivity

☰ Key Hyperparameters

Hyperparameter	Search Range	Optimal
Learning Rate	$\{10^{-5}, 10^{-4}, 10^{-3}\}$	10^{-4}
Batch Size	$\{16, 32, 64, 128\}$	64
Weight Decay	$\{0, 10^{-4}, 10^{-2}\}$	10^{-4}
Dropout	$\{0.0, 0.1, 0.3, 0.5\}$	0.1